Cells and Gauge

File: gauge.kicad_sch

Sleep Current and 3V3

File: power.kicad_sch

MCU and NTCs

File: mcu.kicad_sch

USB Port 1

IRQD IRQ1
EN_5V EN_5V1
ADCIN1 P1_ADCIN1
ADCIN2 P1_ADCIN2
LDO_3V3D P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQD IRQ2
EN_5V EN_5V2
ADCIN1 P2_ADCIN1
ADCIN2 P2_ADCIN2
LDO_3V3D

File: usb_port.kicad_sch

P1_LDO_3V3
R1 0R
P1_ADCIN1
R2 0R
P1_ADCIN2
GND

P2_ADCIN1
R3 0R
P2_ADCIN2
R4 0R
GND

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: R1 ADCIN1 = LDO_3V3, R2 ADCIN2 = GND -> I2C 0x20
Port 2: R3 ADCIN1 = GND,   R4 ADCIN2 = GND -> I2C 0x21
Gauge BQ40Z50-R2 = 0x0B on the same SDA/SCL bus (STM32WB I2C1, 100 kHz)

MONOLITH DEV BOARD - SHEET INDEX
2 Cells and Gauge   4S holders / pack input, BQ40Z50-R2, protection FETs, SMBus
3 Sleep Current and 3V3 R_MEAS + 2x INA190 to SMA, TPS70950 amp supply, TPS62177 3V3
4 MCU and NTCs   STM32WB1MMC, debug headers, board NTCs TH1-TH4, LED
5 USB Port 1 / 6 USB Port 2   (same sheet file) TPS25751D + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (gauge VSS only)
Spec: docs/devboard/circuit-description.md   Changes/open items: DevBoard/README.md

REVISION LOG
A0 2026-10-11 First draft from circuit description draft 5. Author: Claude Code. Reviewer: none yet.
A1 2026-10-11 Layout cleanup: blocks, power symbols, notes, title blocks. Author: Claude Code. Reviewer: none yet.
A2 2026-10-11 Readability pass: no overlapping text, sheet-pin names inside boxes. Author: Claude Code. Reviewer: none yet.
A3 2026-10-11 Wiring pass: wires instead of labels inside each block. Author: Claude Code. Reviewer: none yet.
A4 2026-10-12 Polymer bulk caps, PWR_FLAGs at rail sources, design-review fixes
                (docs/devboard/schematic-review-2026-10-12.md). Author: Claude Code. Reviewer: 3-agent review.
A5 2026-10-13 Review 2 decisions: CHG_INH fails safe (R417/R517 47k pull-up to 3V3, MCU drives low to charge),
                MCU IWDG in hardware mode; BAT fed before F101 (TI fig. 9-12); C421/C521 100 uF; LRB520S, BSS138,
                C109 4.7n, PPHV/VBUS caps, GND/probe test points, JLC numbers on all R/C (review 2). Author: Claude Code.

Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
Sheet index, PD address straps
Monolith 100 W battery bank — dev board

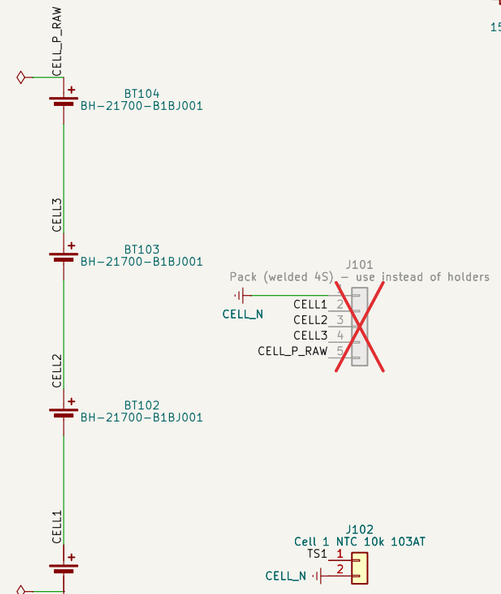
Sheet: /
File: MonolithDev.kicad_sch

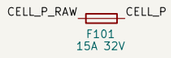
Title: Monolith dev board — top level

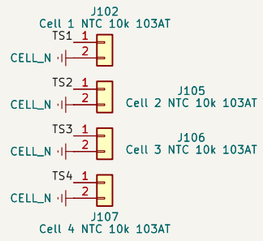
Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 1/6

CELL STACK (4S)

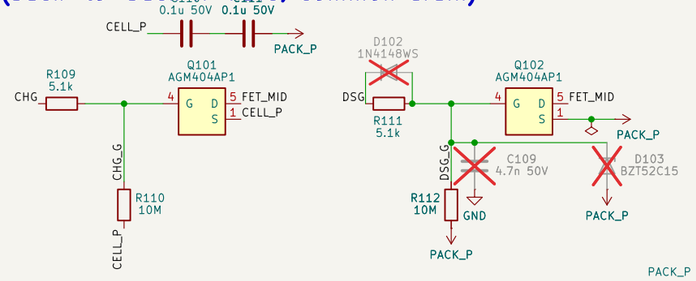
Holders BT101–BT104 OR welded pack on J101 — never both.
J101 is DNP. Check each cell at TP102–TP105 and the stack at TP108 before waking the gauge (Q101/Q102 stay off until FET_EN is set, so nothing loads the cells until then).



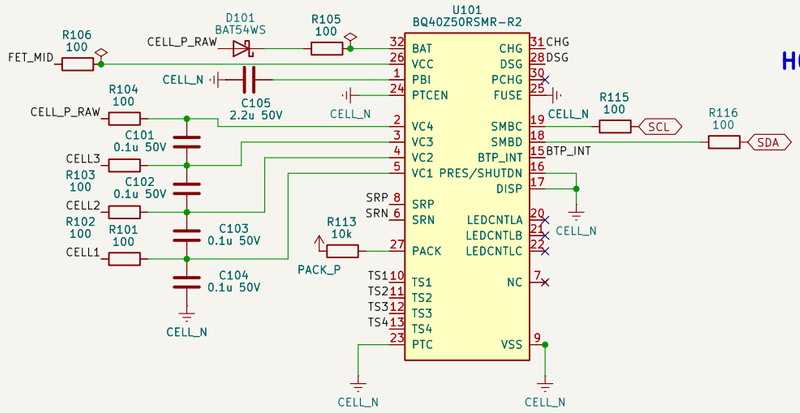




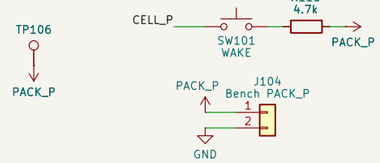
PROTECTION FETs (back-to-back N-FETs, common drain)



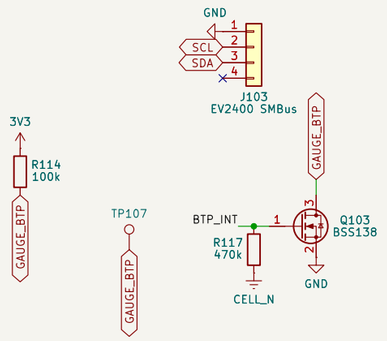
GAS GAUGE BQ40Z50-R2 (TI fig. 9-1)



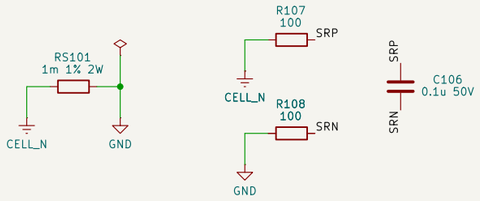
WAKE + BENCH SUPPLY



HOST SMBus (gauge 0x0B) / EV2400



LOW-SIDE CURRENT SENSE



TEST POINTS / POWER FLAGS

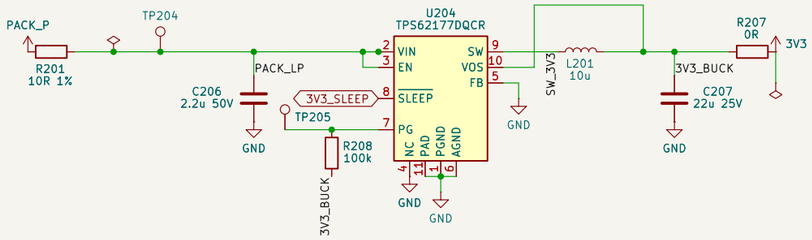


NOTES
1. CELL_N = gauge VSS (cell 1N tab). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads; R107/R108/C106 at the IC.
3. VC taps are separate thin traces from the cell end of each holder contact;
   VC4 from CELL_P_RAW (before F101) so fuse drop is not read as cell voltage.
4. TS1–TS4: one wired cell NTC each (J102, J105–J107, one per cell). PTC, PTCEN, PRES, DISP, FUSE tied to VSS per datasheet.
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_P (after fuse, before FETs) via D101 + R105.
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT (its high level has no maximum; PA6 takes 4 V unpowered): GAUGE_BTP is active low.
9. C109 + D102 (DNP): fit if the DSG turn-on inrush into PACK_P trips SCD. D103 (DNP) clamps Q102 VGS in a hard short.
10. Tape the J102 NTC to BT101: a 21700 can is its cell's negative terminal and TS1 is rated 2.1 V.
11. SW101 wakes the gauge from SHUTDOWN without USB; J104 feeds PACK_P from a current-limited bench supply.
12. J103 matches the EV2400 SMBus port: 1 VSS, 2 SCL, 3 SDA, 4 VOUT (not used). Fit cells before plugging in USB.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2), C107+C108 across PACK+/PACK− (sec. 9.2.2.1.5).

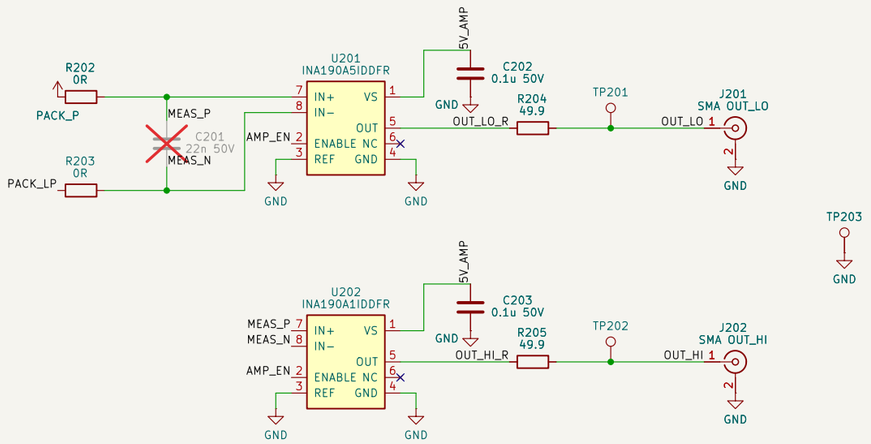
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
4S input, BQ40Z50-R2 gauge/protector, protection FETs
Monolith 100 W battery bank — dev board

Sheet: /Cells and Gauge/		
File: gauge.kicad_sch		
Title: Cells and gauge		
Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 2/6

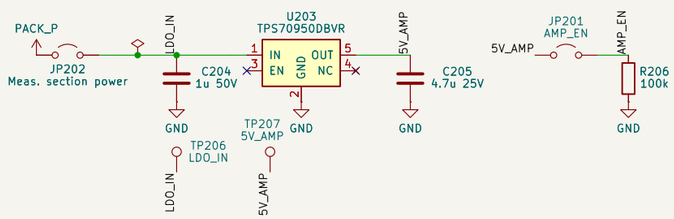
3.3 V ALWAYS-ON RAIL (TPS62177, 1 MHz, sleep mode under MCU control)



SLEEP-CURRENT MEASUREMENT (both amps across R201 = R_MEAS)



AMPLIFIER SUPPLY (5 V LDO, fed from PACK_P so amp current bypasses R_MEAS)



NOTES
1. Low-current branch: PACK_P -> R201 10R -> PACK_LP -> TPS62177 -> 3V3 (MCU + both TPS25751).
   Gauge BAT is NOT on this branch (TI: gauge must stay powered with FETs off).
2. Kelvin-connect R202/R203 to the R201 pads.
3. U201 gain 500: 5 mV per uA, -0.99 mA full scale. U202 gain 25: 0.25 V per mA, -20 mA full scale.
4. R202/R203/C201 fitted 0R/0R/DNP to start; 100R/100R/22nF gives a 36 kHz input filter.
5. R204/R205 49.9R isolate the INA190 outputs from cable capacitance (1 nF max); use a 1 MOhm scope input.
6. SLEEP low = sleep mode (internal pull-down); MCU PB4 drives it high for full load.
7. R207 breaks the 3V3 rail for debugging.
8. U201/U202 are hand-soldered (DigiKey/Mouser); JLC has no stock. Not in the JLC BOM/CPL.
9. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
10. U203 EN is left open (internal 300 nA pull-up enables it). EN must stay below 6.5 V: never tie it to LDO_IN.
11. U204 pin 4 (NC) goes to GND, as TI recommends.

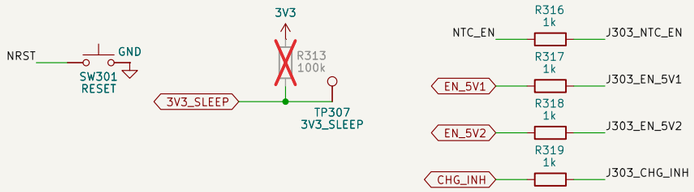
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
R_MEAS + INA190 x2, TPS70950, TPS62177
Monolith 100 W battery bank — dev board

Sheet: /Sleep Current and 3V3/
File: power.kicad_sch

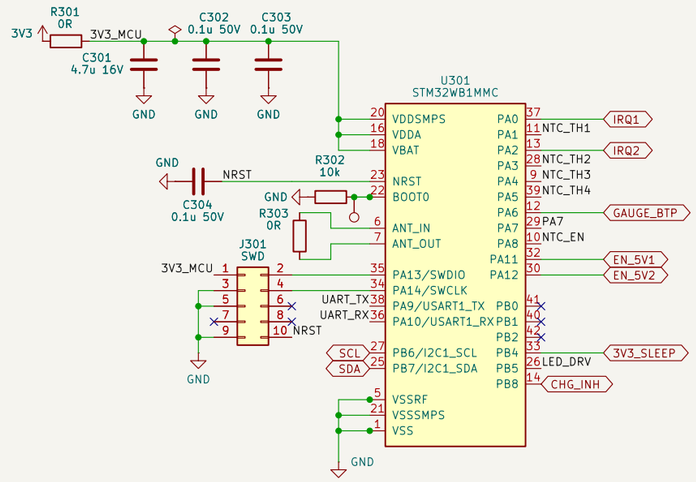
Title: Sleep-current measurement and 3.3 V rail

Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 3/6

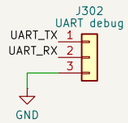
BRING-UP AIDS



BLUETOOTH MCU STM32WB1MMC (per ST DS14096 fig. 5)

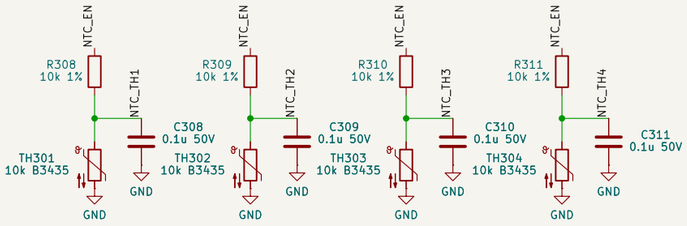


DEBUG HEADERS

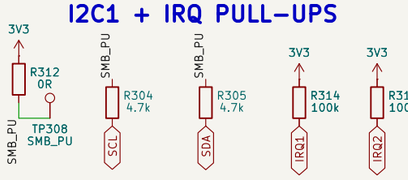




BOARD NTCs (TH1–TH4, divider powered only while sampling)

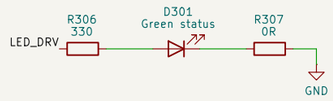


I2C1 + IRQ PULL-UPS



NOTES
1. U301 at a board edge, as on ST's reference PCB (DS14096 fig. 6): copper-free notch past the antenna end.
2. C301–C303 at the VDD pads; C304 at NRST.
3. PB0/PB1/PB2 left unconnected (ST: RF-sensitive pins). NTC_EN = PA8, EN_5V1 = PA11, EN_5V2 = PA12.
4. R303 0R joins ANT_IN/ANT_OUT for the internal antenna (ST fig. 2).
5. Pin map checked against DS13258 table 15: NTC ADC1_IN6/8/9/10, IRQs on WKUP1/WKUP4, I2C1, USART1.
   PB4 = 3V3_SLEEP: PB4 resets as NJTRST with pull-up, so the 3V3 rail boots in normal mode;
   firmware makes PB4 a GPIO output before using sleep mode. PB5 = LED, PA6 = GAUGE_BTP.
6. R307 removes the LED for sleep-current measurements.
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. SW301 resets the MCU. TP306 is BOOT0: hold it at 3V3 while releasing reset for the UART bootloader.
9. R313 (DNP) holds 3V3_SLEEP high (normal mode) while the MCU is off. TP307 probes it.
10. R312 feeds the SMBus pull-ups: remove it and supply TP308 for EV2400 work while 3V3 is down.
11. R316–R319: J303 can override NTC_EN/EN_5V1/EN_5V2/CHG_INH without shorting the MCU outputs.
12. PB8 = CHG_INH (rev A5): drive LOW to allow charging. High or hi-Z stops both chargers (R417/R517 pull CE up to 3V3), so reset/hang/unprogrammed stops charging. Low only while charging (69 uA per port), high in idle. No Standby.
13. Watchdog: IWDG in hardware mode (option byte IWDG_SW = 0) restarts a hung MCU; it runs from LSI, also in Stop/Standby.

STATUS LED





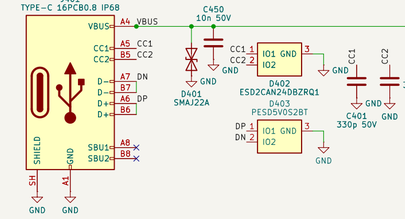
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
STM32WB1MMC, debug headers, NTC TH1–TH4
Monolith 100 W battery bank — dev board

Sheet: /MCU and NTCs/
File: mcu.kicad_sch

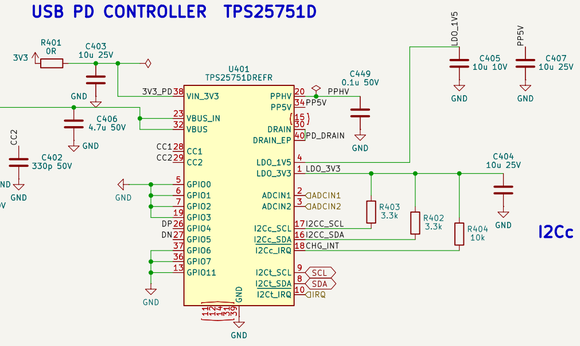
Title: MCU and board NTCs

Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 4/6

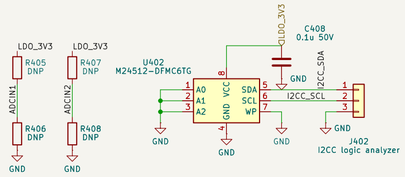
USB-C CONNECTOR + PROTECTION



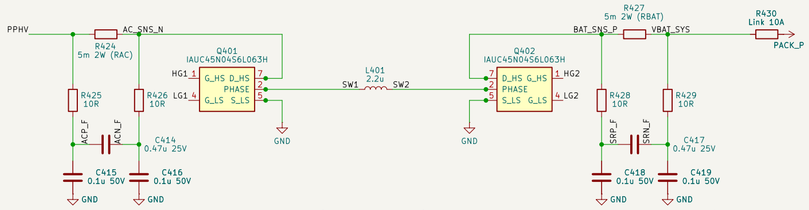
USB PD CONTROLLER TPS25751D



I2Cc BUS: EEPROM + PULL-UPS

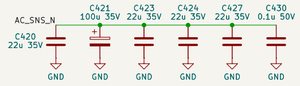


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)



USB-SIDE BULK: 4 x 22u MLCC + 68u polymer (>= 80 uF effective)

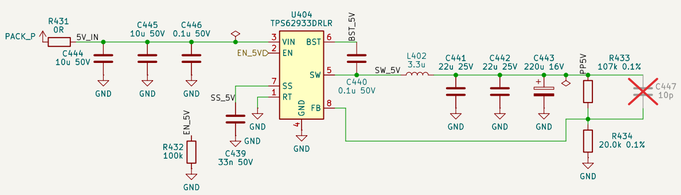
BATTERY-SIDE BULK: 3 x 47u MLCC + 100u polymer (>= 80 uF effective)



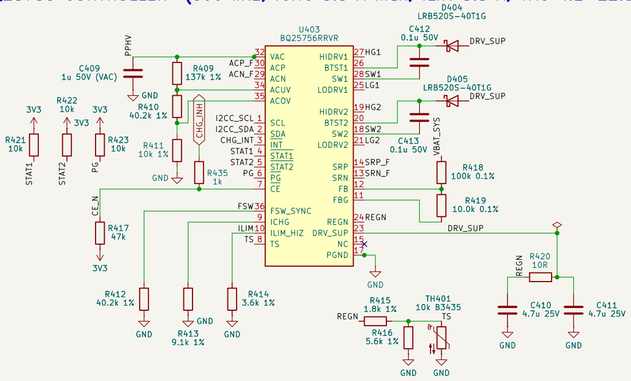


NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). R405–R408 are DNP divider footprints for other options.
2. Unused TPS25751 GPIOs tied to GND per datasheet (KiCad ERC flags this; expected).
3. At the connector pins: D402 CC ESD (24 V). D403 D+/D− ESD (5 V; GPIO4/5 are rated 6 V). C401/C402.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the polymer (C421/C432) and remaining MLCCs one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. R430 = 10 A copper link to PACK_P (net-tie footprint, not a part): widen with zones.
8. VAC is taken before R424 per TI typical application: DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5xxBulkPd: the TPS25751D sink switch soft-start limits inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0). Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH (rev A5): R417 pulls CE up to 3V3, so charging needs the MCU to drive CHG_INH low through R435. Reset, IWDG reset, a hang or an unprogrammed MCU stop charging; a dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.

PP5V SUPPLY TPS62933 (1.2 MHz, 5.00 V) - 5 V source + VCONN



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.5 A, VAC 4.1–22.5 V window)



TEST POINTS / POWER FLAGS



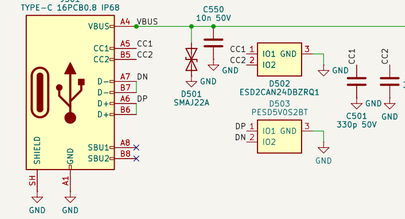
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — dev board

Sheet: /USB Port 1/
File: usb_port.kicad_sch

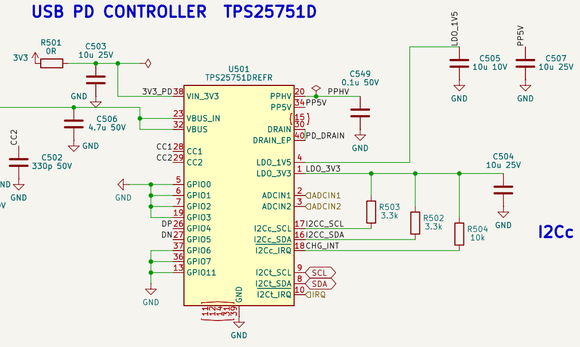
Title: USB-C PD port (used twice)

Size: A2	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 5/6

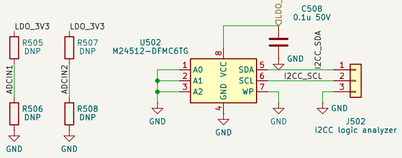
USB-C CONNECTOR + PROTECTION



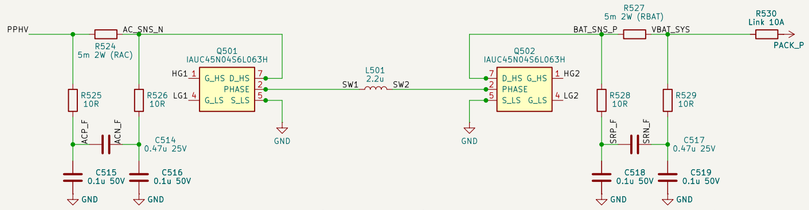
USB PD CONTROLLER TPS25751D



I2Cc BUS: EEPROM + PULL-UPS

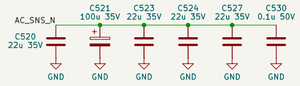


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)



USB-SIDE BULK: 4 x 22u MLCC + 68u polymer (>= 80 uF effective)

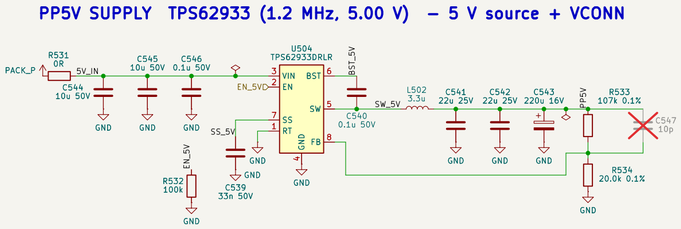
BATTERY-SIDE BULK: 3 x 47u MLCC + 100u polymer (>= 80 uF effective)



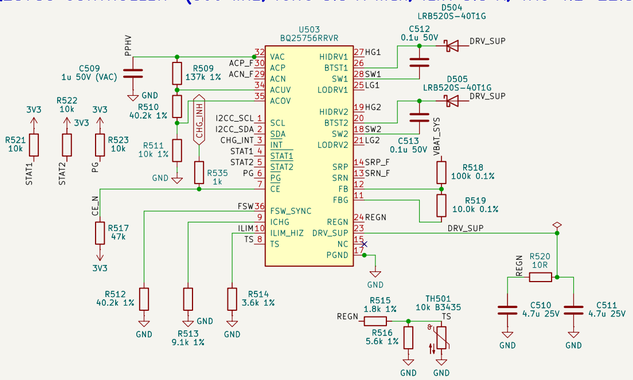


NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). R405–R408 are DNP divider footprints for other options.
2. Unused TPS25751 GPIOs tied to GND per datasheet (KiCad ERC flags this; expected).
3. At the connector pins: D402 CC ESD (24 V). D403 D+/D− ESD (5 V; GPIO4/5 are rated 6 V). C401/C402.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the polymer (C421/C432) and remaining MLCCs one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. R430 = 10 A copper link to PACK_P (net-tie footprint, not a part): widen with zones.
8. VAC is taken before R424 per TI typical application: DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5xxBulkPd: the TPS25751D sink switch soft-start limits inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0). Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH (rev A5): R417 pulls CE up to 3V3, so charging needs the MCU to drive CHG_INH low through R435. Reset, IWDG reset, a hang or an unprogrammed MCU stop charging; a dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.

PP5V SUPPLY TPS62933 (1.2 MHz, 5.00 V) - 5 V source + VCONN



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.5 A, VAC 4.1–22.5 V window)



TEST POINTS / POWER FLAGS



Not reviewed, laid out, built or tested	
Spec + open items: DevBoard/README.md	
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx	
Monolith 100 W battery bank — dev board	
Sheet: /USB Port 2/	
File: usb_port.kicad_sch	
Title: USB-C PD port (used twice)	
Size: A2	Date: 2026-10-11
KiCad E.D.A. 10.0.7	**Rev: A (draft)**
	Id: 6/6